

Fixed subspaces and reconstruction of the generators

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Scope

Decreasing transported projections determine common fixed subspaces by strong vectorwise limits. Shell sums and residual corners reconstruct the generators in arbitrary unital target representations. For a nonzero irreducible target representation, a common fixed space survives and supplies compatible selected vectors.

8 direct Cards + 12 reused prerequisites = 20 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Reconstructing a represented unitary from its shells and residual corner
- Key result: An irreducible target representation has a surviving fixed space

[Read this route with prerequisites](#)

Detailed route conditions

For the supplied family, write $f_{i,n} = \alpha_i(f_n)$. On the matching fiber its common fixed projection is the projection onto $\mathbb{C}\xi_i$; on an inequivalent fiber it is zero. Coordinatewise reasoning therefore identifies the common range in $\mathcal{H}$ with $\mathbb{C}b_i$. These are strong, vectorwise consequences of decreasing projections, not operator-norm limits.

In an arbitrary representation of the generated algebra, shell sums and their adjoints are reconstructed on that representation's own Hilbert space. The generator splits into a strong shell sum and its residual corner. Nonzero irreducibility prevents every transported fixed space from vanishing. A surviving index i_0 may be different from the root class o ; its vector is transported to a common root vector ζ , then pulled back to the compatible vectors η_i . No equality $\eta_o = \zeta$ is imposed until the additional root-generator normalization is available.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (5 Cards)

Level 0 The selected GNS representation of a pure-state class A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class. MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation Immediate Card prerequisites: None in this selected scope Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations Open Declaration Card Read exact source
Level 0 The CAR algebra as a completion of finite matrix stages Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation. MathlibAnnex.CStarAlgebra.CAR.Limit Immediate Card prerequisites: None in this selected scope Used by in this scope: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra Open Declaration Card Read exact source
Level 0 A fixed representative family of CAR shell data One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component. MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily Immediate Card prerequisites: None in this selected scope Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Shell matching fixes the trace of transported flags · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source

Level 0 The closed operator algebra generated by a representation and extra operators Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra. MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget Immediate Card prerequisites: None in this selected scope Used by in this scope: Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 0 The normalized trace on a finite CAR stage The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions. MathlibAnnex.CStarAlgebra.CAR.stageTrace Immediate Card prerequisites: None in this selected scope Used by in this scope: Constructing the normalized trace on the completed CAR algebra Open Declaration Card Read exact source
Level 1 (4 Cards)
Level 1 Constructing the normalized trace on the completed CAR algebra Carries compatible bounded matrix traces through the algebraic limit, normed union and completion. MathlibAnnex.CStarAlgebra.CAR.trace Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages Used by in this scope: Shell matching fixes the trace of transported flags Open Declaration Card Read exact source
Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. MathlibAnnex.CStarAlgebra.CAR.rootState Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state Open Declaration Card Read exact source

Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range Open Declaration Card Read exact source
Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Shell matching fixes the trace of transported flags · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 2 (3 Cards) Level 2 Shell matching fixes the trace of transported flags Recovers exact trace values from the initial and final supports of shell links. MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag Immediate Card prerequisites: The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · A fixed representative family of CAR shell data Used by in this scope: Transported flags vanish on vectors generated by a trace vector Open Declaration Card Read exact source Level 2 Root compression becomes scalar in norm Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra. MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · A common root vector realizes every selected state through the generators Open Declaration Card Read exact source

Level 2 Purity of the completed root state Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra. <code>MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState</code> Immediate Card prerequisites: The product-vector state on the completed CAR algebra Used by in this scope: The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range Open Declaration Card Read exact source
Level 3 (4 Cards)
Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne</code> Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source
Level 3 Transported flags vanish on vectors generated by a trace vector Turns decay of projection traces into norm convergence on each source-orbit vector. <code>MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero</code> Immediate Card prerequisites: Shell matching fixes the trace of transported flags Used by in this scope: None in this selected scope Open Declaration Card Read exact source

Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero</code> Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source
Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. <code>MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation</code> Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 4 (2 Cards)
Level 4 The atomic common range is one embedded GNS line Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum. <code>MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span</code> Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range Used by in this scope: None in this selected scope Open Declaration Card Read exact source
Level 4 Reconstructing a represented unitary from its shells and residual corner Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces. <code>MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction</code> Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators Used by in this scope: An irreducible target representation has a surviving fixed space Open Declaration Card Read exact source

Level 5 (1 Card)

Level 5
An irreducible target representation has a surviving fixed space
Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.
<code>MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace</code>
Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner
Used by in this scope: A common root vector realizes every selected state through the generators
Open Declaration Card Read exact source

Level 6 (1 Card)

Level 6
A common root vector realizes every selected state through the generators
Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states.
<code>MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily</code>
Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space
Used by in this scope: None in this selected scope
Open Declaration Card Read exact source